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Course: Query Processing and Data Science

Code: DSA0502

CO1 – ASSIGNMENT TOOL 2

1. Employee Data Cleaning (CSV)

A company stores employee information in employees.csv. Some employee names contain extra spaces, salaries are stored as text, and a few records have missing values.

Task:

- Read the CSV file.
- Remove extra spaces from employee names.
- Convert salary into numeric format.
- Fill missing salary values with the average salary.
- Save the cleaned dataset as cleaned_employees.csv.

ANSWER:

Employees.csv DATASET

EmployeeID	Name	Department	Salary
101	" Alice Johnson "	HR	45000
102	"Bob Smith"	Finance	52000
103	" Charlie Brown"	IT	60000
104	"Diana Prince "	Marketing	48000
105	" Ethan Hunt "	Sales	
106	"Fiona Green"	HR	47000
107	" George Miller "	IT	abc
108	"Helen White"	Finance	55000
109	" Ian Black"	Sales	51000
110	"Julia Adams "	Marketing	
111	" Kevin Lee "	IT	62000
112	"Laura Scott"	HR	46000
113	" Michael King "	Finance	53000
114	"Nancy Hall"	Sales	49000
115	" Oliver Stone "	Marketing	58000
116	"Patricia Clark"	IT	
117	" Quentin Ross "	HR	44000

118,"Rachel Evans",Finance,56000
119," Steven Young ",Sales,50000
120," Tina Turner ",Marketing,57000

PROGRAM:

```
CO1 > AT2 > co1-at2-1.py > ...  
1 import pandas as pd  
2 df = pd.read_csv(r"C:\Users\dhars\Documents\Query processing\lab_questions\CO1\AT2\co1_at2.csv")  
3 df["Name"] = df["Name"].str.strip()  
4 df["Salary"] = pd.to_numeric(df["Salary"], errors="coerce")  
5 average_salary = df["Salary"].mean()  
6 df["Salary"] = df["Salary"].fillna(average_salary)  
7 df.to_csv("cleaned_employees.csv", index=False)  
8 print("Employee data cleaned successfully.")  
9 print(df)
```

OUTPUT:

Employee data cleaned successfully.

	EmployeeID	Name	Department	Salary
0	101	Alice Johnson	HR	45000.0
1	102	Bob Smith	Finance	52000.0
2	103	Charlie Brown	IT	60000.0
3	104	Diana Prince	Marketing	48000.0
4	105	Ethan Hunt	Sales	52062.5
5	106	Fiona Green	HR	47000.0
6	107	George Miller	IT	52062.5
7	108	Helen White	Finance	55000.0
8	109	Ian Black	Sales	51000.0
9	110	Julia Adams	Marketing	52062.5
10	111	Kevin Lee	IT	62000.0
11	112	Laura Scott	HR	46000.0
12	113	Michael King	Finance	53000.0
13	114	Nancy Hall	Sales	49000.0
14	115	Oliver Stone	Marketing	58000.0
15	116	Patricia Clark	IT	52062.5
16	117	Quentin Ross	HR	44000.0
17	118	Rachel Evans	Finance	56000.0
18	119	Steven Young	Sales	50000.0
19	120	Tina Turner	Marketing	57000.0

2. Student Marks Analysis

A college maintains student marks in students.csv.

Task:

- Read the dataset.
- Calculate the average marks.
- Identify students who scored below 40.
- Add a new column named Result (Pass/Fail).
- Export the updated file.

ANSWER:

Students.csv DATASET:

StudentID,Name,Department,Marks

201,Aarav Sharma,CSE,85

202,Diya Patel,ECE,72

203,Rohan Gupta,EEE,39

204,Meera Nair,CSE,91

205,Arjun Singh,MECH,28

206,Sneha Reddy,CIVIL,67

207,Karan Verma,CSE,45

208,Priya Iyer,ECE,33

209,Rahul Das,EEE,58

210,Ananya Roy,CSE,76

211,Vikram Kumar,MECH,40

212,Neha Joshi,CIVIL,37

213,Aditya Rao,ECE,81

214,Kavya Menon,CSE,95

215,Manoj Babu,EEE,22

216,Pooja Shah,MECH,49

217,Sanjay Mishra,CIVIL,60

218,Ishita Kapoor,ECE,18

219,Nitin Yadav,CSE,55

220,Swathi Krishnan,EEE,88

PROGRAM:

```
CO1 > AT2 > co1-at2-2.py > ...
1  import pandas as pd
2  df = pd.read_csv(r"C:\Users\dhars\Documents\Query processing\lab_questions\CO1\AT2\co1_at2_2.csv")
3  average_marks = df["Marks"].mean()
4  print("Average Marks:", average_marks)
5  failed_students = df[df["Marks"] < 40]
6  print("\nStudents scoring below 40:")
7  print(failed_students)
8  df["Result"] = df["Marks"].apply(lambda x: "Pass" if x >= 40 else "Fail")
9  df.to_csv("updated_students.csv", index=False)
10 print("\nUpdated student data saved successfully.")
11 print(df)
```

OUTPUT:

Average Marks: 56.95

Students scoring below 40:

	StudentID	Name	Department	Marks
2	203	Rohan Gupta	EEE	39
4	205	Arjun Singh	MECH	28
7	208	Priya Iyer	ECE	33
11	212	Neha Joshi	CIVIL	37
14	215	Manoj Babu	EEE	22
17	218	Ishita Kapoor	ECE	18

Updated student data saved successfully.

	StudentID	Name	Department	Marks	Result
0	201	Aarav Sharma	CSE	85	Pass
1	202	Diya Patel	ECE	72	Pass
2	203	Rohan Gupta	EEE	39	Fail
3	204	Meera Nair	CSE	91	Pass
4	205	Arjun Singh	MECH	28	Fail
5	206	Sneha Reddy	CIVIL	67	Pass
6	207	Karan Verma	CSE	45	Pass
7	208	Priya Iyer	ECE	33	Fail
8	209	Rahul Das	EEE	58	Pass
9	210	Ananya Roy	CSE	76	Pass
10	211	Vikram Kumar	MECH	40	Pass
11	212	Neha Joshi	CIVIL	37	Fail
12	213	Aditya Rao	ECE	81	Pass
13	214	Kavya Menon	CSE	95	Pass
14	215	Manoj Babu	EEE	22	Fail
15	216	Pooja Shah	MECH	49	Pass
16	217	Sanjay Mishra	CIVIL	60	Pass
17	218	Ishita Kapoor	ECE	18	Fail
18	219	Nitin Yadav	CSE	55	Pass
19	220	Swathi Krishnan	EEE	88	Pass